SHANMUKHA CHATADI

+1 (984)-381-0808 | schatad@ncsu.edu | shanmukha-chatadi | github.com/shinegami-2002

SUMMARY

Applied AI/ML Engineer building **agentic AI systems**, production LLM pipelines, and scalable cloud deployments. Architected enterprise RAG platform serving **230+ users** at **75% lower cost**; shipped multi-agent **LangGraph** research assistant with corrective RAG and hallucination detection. Published researcher (Springer, IEEE) in **deep learning, NLP**, and **computer vision**.

EDUCATION

North Carolina State University — Raleigh, NC
Master of Science in Computer Science *(Data Science Track)*, **3.71** — Aug 2024 – Jun 2026

National Institute of Technology Puducherry — Pondicherry, India
Bachelor of Technology in Computer Science and Engineering, **8.48** — Dec 2020 – Mar 2024

Coursework: Deep Learning, Machine Learning, Neural Networks, Algorithms, Software Engineering

TECHNICAL SKILLS

AI/ML	RAG, Agentic RAG, LLMs, NLP, Agentic AI, Multi-Agent Systems, MCP, Deep Learning, CNNs, Transformers, Computer Vision, Fine-tuning, Transfer Learning, HITL, MLOps
Languages	Python, TypeScript, JavaScript, Go, SQL, C++, Java, Bash
Libraries	LangGraph, LangChain, AWS Bedrock, ChromaDB, OpenSearch k-NN, Hugging Face, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, XGBoost, spaCy, Matplotlib
Cloud & DevOps	AWS (Bedrock, SageMaker, Lambda, S3, OpenSearch, EKS, EC2, API Gateway, Cognito), Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD, Git
Web & APIs	FastAPI, Next.js, React, Flask, PostgreSQL, MongoDB, Redis, Vector Databases, GraphQL, REST APIs

WORK EXPERIENCE

AI Research Intern, Michigan Health Information Network (MiHIN), Lansing, MI — Jul 2025 – Present

- Architected **production Document AI/RAG platform** on **AWS Bedrock**, OpenSearch Serverless (k-NN vector search), and Lambda, enabling **230+ employees** to query **1T+ documents** via natural language at **75% less than competitors**.
- Engineered **NLP ingestion pipeline** with document parsing, **Titan V2 embedding** generation (1,024-dim), configurable chunking, vector indexing, and retrieval; incorporated **human-in-the-loop (HITL)** feedback to tune chunk size, overlap, and top-k parameters, reducing LLM hallucination rates by **30%**.
- Developed **AI governance framework** aligned with **HITRUST CSF, NIST, HIPAA**; spearheaded AI Task Force training across DevOps, Legal, and HR, authoring playbooks that enabled **5+ cross-functional teams** to integrate generative AI into production workflows.
- Delivered serverless REST API with **API Gateway, Lambda, Cognito** (SSO + 2FA) supporting **230+ users**; presented cost analysis to **CEO**, securing approval for org-wide pilot cutting costs by **75%**.

Data & Software Development Analyst, Rygen BioPharma, Hyderabad, India — Mar 2023 – Aug 2024

- Built **end-to-end ML pipelines** with **Python** (scikit-learn, Pandas, NumPy) for clinical outcome prediction: performed **feature engineering**, trained classification and regression models (XGBoost, Random Forest) with hyperparameter tuning (GridSearchCV), achieving **0.89 AUC-ROC** across **100K+ records**.
- Designed **ETL/data pipelines** processing **100K+ records daily** with schema validation and feature extraction; standardized ingestion from **4+ departments**, accelerating data delivery by **50%**. Built data quality system with **PostgreSQL** triggers, reducing errors by **50%**.
- Deployed **containerized microservices** for clinical APIs on **AWS** with **Docker and Kubernetes**; implemented **CI/CD pipelines with Jenkins**, reducing release cycles by **35%**.

Data Science Intern, TreoSoft IT Solutions, Bengaluru, India — Jun 2022 – Sep 2022

- Shipped **ML-powered recommendation engine** with **Flask REST APIs and PostgreSQL**; applied Apriori/FP-Growth association mining and K-Means segmentation on **100K+ transactions**, driving **15% increase in net sales** and **35% revenue uplift**.

PROJECTS

ScholarAgent – *LangGraph, ChromaDB, Gemini API, FastAPI, Next.js 14, TypeScript* [GitHub] — Jan 2025 – Mar 2025

- Orchestrated **multi-agent research assistant** with **7-node LangGraph pipeline** (router, retriever, grader, rewriter, generator, hallucination checker, synthesizer); implemented **corrective RAG** with query rewriting and hallucination detection, improving groundedness by **35%**.
- Integrated **ChromaDB** vector search with **HuggingFace sentence-transformer** embeddings indexing **10K+ arXiv/PubMed papers**; constructed **3-model Gemini cascade** achieving **99% uptime** and Next.js 14 + TypeScript frontend.

MCP Healthcare Server – *Python, FastMCP, httpx, openFDA/PubMed/NLM/CMS APIs, Docker* [GitHub] — Feb 2025 – Mar 2025

- Launched **Model Context Protocol server** exposing **6 healthcare tools** (drug safety, clinical trials, ICD-10 lookup, medical literature, Medicare data, drug interactions) via **FastMCP**; integrated **4 federal APIs** with async httpx achieving sub-second response times. Wrote **21 unit tests**, CI across Python 3.11–3.13 matrix.

Neural Network Training Dynamics & Pruning – *PyTorch, UMAP/t-SNE, ResNet-18, ViT-Tiny* — Jan 2026 – Present

- Trained **ResNet-18 and ViT-Tiny** on CIFAR-100 using **transfer learning**; analyzed CNN vs. Transformer representation learning via **UMAP/t-SNE** and CKA similarity. Applied **50% L1 unstructured pruning** – pruned ViT retained **92% top-5 accuracy** vs. 3% drop in pruned ResNet, quantifying model compression trade-offs.

Publications: C. Shanmukha Srinivas Sai et al., *Potato Disease Classification Using ML Models*, HIS 2023 (Springer) | **Somesh K, C. Shanmukha Srinivas Sai** et al., *Tea Leaf Disease Classification with Ensemble Stacking*, ICDLAIR 2024 (IEEE)